Buổi 7: Ôn các bài toán nâng cao GKI

Trích đề cương GKI trường THCS Thành Công

C – Một số bài toán nâng cao:

Bài 15. Tìm số nguyên n biết: https://youtu.be/lLL4fh5Nwi4

a) $\frac{343}{125}=\left(\frac{7}{5}\right)^n$

$\frac{7^3}{5^3}=\left(\frac{7}{5}\right)^n$ $\quad \frac{a^n}{b^n}=\left(\frac{a}{b}\right)^n$

$\left(\frac{7}{5}\right)^3=\left(\frac{7}{5}\right)^n$

$n=3\in\mathbb{Z}$

Vậy $n=3$

b) $5^{3n-2}=625$

$5^{3n-2}=5^4$

$3n-2=4$

$3n=6$

$n=2\in\mathbb{Z}$

Vậy: $n=2$

c) $3^{n+1}-2=5^2$

$3^{n+1}=27$

$3^{n+1}=3^3$

$n+1=3$

$n=2\in\mathbb{Z}$

Vậy $n=2$

d) $\frac{-3}{4}<\frac{n}{10}<\frac{-3}{5}$

$\frac{-15}{20}<\frac{2n}{20}<\frac{-12}{20}$

$-15<2n<-12$

Chia các vế cho 2 ta được:

$-7,5<n<-6$

Vì $n\in\mathbb{Z}$ nên $n=-7$

e) $\frac{-5}{12}<\frac{n}{5}<\frac{1}{4}$ https://youtu.be/sRHphs0JHE4

$\frac{-25}{60}<\frac{12n}{60}<\frac{15}{60}$

$-25<12n<15$

Vì $n\in\mathbb{Z}$ nên ta có bảng giá trị:

n	-4	-3	-2	-1	0	1	2
$12n$	-48	-36	-24	-12	0	12	24

Vậy: $n\in\{-2;-1;0;1\}$

Bài 16. Tính giá trị các biểu thức sau:

$\frac{1}{6}=\frac{1}{3}\cdot\frac{1}{2}$

$M=\dfrac{\frac{2}{3}-\frac{2}{5}-\frac{2}{7}+\frac{2}{11}}{\frac{13}{3}-\frac{13}{5}-\frac{13}{7}+\frac{13}{11}}=\dfrac{2\cdot\left(\frac{1}{3}-\frac{1}{5}-\frac{1}{7}+\frac{1}{11}\right)}{13\cdot\left(\frac{1}{3}-\frac{1}{5}-\frac{1}{7}+\frac{1}{11}\right)}=\frac{2}{13}$

$N=\dfrac{\frac{1}{6}-\frac{1}{39}+\frac{1}{51}}{\frac{1}{8}-\frac{1}{52}+\frac{1}{68}}=\dfrac{\frac{1}{3}\cdot\frac{1}{2}-\frac{1}{3}\cdot\frac{1}{13}+\frac{1}{3}\cdot\frac{1}{17}}{\frac{1}{4}\cdot\frac{1}{2}-\frac{1}{4}\cdot\frac{1}{13}+\frac{1}{4}\cdot\frac{1}{17}}$

$=\dfrac{\frac{1}{3}\cdot\left(\frac{1}{2}-\frac{1}{13}+\frac{1}{17}\right)}{\frac{1}{4}\cdot\left(\frac{1}{2}-\frac{1}{13}+\frac{1}{17}\right)}=\frac{1}{3}:\frac{1}{4}=\frac{1}{3}\cdot\frac{4}{1}=\frac{4}{3}$

$P=\dfrac{2^{13}\cdot 3}{2^{15}\cdot 3^2\cdot 9^2}=\frac{1}{2^2\cdot 3\cdot 81}=\frac{1}{4\cdot 3\cdot 81}=\frac{1}{972}$

$Q=\dfrac{4^5\cdot 9^4-2\cdot 6^9}{2^{10}\cdot 3^8+6^8\cdot 20}=\dfrac{2^{10}\cdot 3^8-2\cdot 2^9\cdot 3^9}{2^{10}\cdot 3^8+2^8\cdot 3^8\cdot 2^2\cdot 5}$

$4^5=(2^2)^5=2^{10}$

$6^9=(2\cdot 3)^9=2^9\cdot 3^9$

$20=2^2\cdot 5$

$=\dfrac{2^{10}\cdot 3^8-2^{10}\cdot 3^8}{2^{10}\cdot 3^8+2^{10}\cdot 3^8\cdot 5}$

$=\dfrac{2^{10}\cdot 3^8\cdot(1-3)}{2^{10}\cdot 3^8\cdot(1+5)}$

$=\frac{1-3}{1+5}=\frac{-2}{6}=-\frac{1}{3}$

https://youtu.be/auTmxioOzJg

$\frac{2}{n.(n+2)} = \frac{1}{n} - \frac{1}{n+2}$; $\frac{a}{n.(n+a)} = \frac{1}{n} - \frac{1}{n+a}$

$A = \frac{4}{5\cdot 7} + \frac{4}{7\cdot 9} + \ldots + \frac{4}{59\cdot 61}$

$= 2\cdot\left(\frac{2}{5.7} + \frac{2}{7.9} + \ldots + \frac{2}{59.61}\right)$

$= 2\cdot\left(\frac{1}{5} - \frac{1}{7} + \frac{1}{7} - \frac{1}{9} + \ldots + \frac{1}{59} - \frac{1}{61}\right)$

$= 2\cdot\left(\frac{1}{5} - \frac{1}{61}\right) = 2\cdot\left(\frac{61}{305} - \frac{5}{305}\right)$

$= 2\cdot\frac{56}{305} = \frac{112}{305}$

$B = \left(1-\frac{2}{5}\right)\left(1-\frac{2}{7}\right)\left(1-\frac{2}{9}\right)\ldots\left(1-\frac{2}{99}\right)$

$= \frac{3}{5}\cdot\frac{5}{7}\cdot\frac{7}{9}\cdot\ldots\cdot\frac{95}{97}\cdot\frac{97}{99}$ ($1-\frac{2}{97}$)

$= \frac{3}{99} = \frac{1}{33}$

Bài 17. So sánh hai biểu thức:

a) 8^{12} và 32^6

$8^{12} = (2^3)^{12} = 2^{36}$

$32^6 = (2^5)^6 = 2^{30}$

So sánh: $2^{36} > 2^{30}$

Vậy $8^{12} > 32^6$

b) 3^{4000} và 9^{2000}

c) $A = \frac{2008^{2008}+1}{2008^{2009}+1}$ và $B = \frac{2008^{2007}+1}{2008^{2008}+1}$ https://youtu.be/zNLqAQa4Ae8

$A = \frac{2008^{2008}+1}{2008^{2009}+1} \Rightarrow 2008A = 2008\cdot\left(\frac{2008^{2008}+1}{2008^{2009}+1}\right)$

$2008A = \frac{2008^{2009}+2008}{2008^{2009}+1} = \frac{2008^{2009}+1+2007}{2008^{2009}+1}$

$\frac{n+m}{n} = \frac{n}{n} + \frac{m}{n}$

$2008A = 1 + \frac{2007}{2008^{2009}+1}$

$B = \frac{2008^{2007}+1}{2008^{2008}+1} \Rightarrow 2008B = \frac{2008.(2008^{2007}+1)}{2008^{2008}+1} = \frac{2008^{2008}+2008}{2008^{2008}+1}$

$2008B = \frac{2008^{2008}+1+2007}{2008^{2008}+1}$

$2008B = 1 + \frac{2007}{2008^{2008}+1}$

So sánh: 2008A và 2008B

$\frac{2007}{2008^{2009}+1} < \frac{2007}{2008^{2008}+1} \Rightarrow 2008A < 2008B$

Vậy: $A < B$

Bài 18. Tính giá trị của x, biết https://youtu.be/Odz5NyoWUXE

a) $\frac{1909-x}{91}+\frac{1907-x}{93}+\frac{1905-x}{95}+\frac{1903-x}{97}+4=0$ https://youtu.be/Y-Qygok-px0

$\left(\frac{1909-x}{91}+1\right)+\left(\frac{1907-x}{93}+1\right)+\left(\frac{1905-x}{95}+1\right)+\left(\frac{1903-x}{97}+1\right)=0$

$\frac{2000-x}{91}+\frac{2000-x}{93}+\frac{2000-x}{95}+\frac{2000-x}{97}=0$

$(2000-x).\left(\frac{1}{91}+\frac{1}{93}+\frac{1}{95}+\frac{1}{97}\right)=0$

$\Rightarrow 2000-x=0$ vì $\frac{1}{91}+\frac{1}{93}+\frac{1}{95}+\frac{1}{97}\neq 0$

$x=2000$

b) $\frac{x+1}{65}+\frac{x+3}{63}=\frac{x+5}{61}+\frac{x+7}{59}$

$\left(\frac{x+1}{65}+1\right)+\left(\frac{x+3}{63}+1\right)=\left(\frac{x+5}{61}+1\right)+\left(\frac{x+7}{59}+1\right)$

$\frac{x+66}{65}+\frac{x+66}{63}=\frac{x+66}{61}+\frac{x+66}{59}$

$a=b$

$a-b=0$

$\frac{x+66}{65}+\frac{x+66}{63}-\frac{x+66}{61}-\frac{x+66}{59}=0$

$(x+66).\left(\frac{1}{65}+\frac{1}{63}-\frac{1}{61}-\frac{1}{59}\right)=0$

$\Rightarrow x+66=0$ $\left(\text{vì } \frac{1}{65}+\frac{1}{63}-\frac{1}{61}-\frac{1}{59}\neq 0\right)$

$x=-66$.

c) $\frac{x+29}{31}-\frac{x+27}{33}=\frac{x+17}{43}-\frac{x+15}{45}$ d) $\frac{1}{x(x+1)}+\frac{1}{(x+1)(x+2)}+\frac{1}{(x+2)(x+3)}-\frac{1}{x}=\frac{1}{2010}$

$\frac{x+29}{31}+\frac{x+15}{45}=\frac{x+17}{43}+\frac{x+27}{33}$ https://youtu.be/j-vgOGtMB7U

Làm tương tự câu b) https://youtu.be/IQ4jDg60ral

c) $\frac{x+29}{31}-\frac{x+27}{33}=\frac{x+17}{43}-\frac{x+15}{45}$ https://youtu.be/SvRN4kY4gN4

$\frac{x+29}{31}+1-\frac{x+27}{33}-1=\frac{x+17}{43}+1-\frac{x+15}{45}-1$

$\left(\frac{x+29}{31}+1\right)-\left(\frac{x+27}{33}+1\right)=\left(\frac{x+17}{43}+1\right)-\left(\frac{x+15}{45}+1\right)$

$-\frac{x+15}{45}\oplus 1$

$-\frac{x+15}{45}+\frac{45}{45}$

$=-\left(\frac{x+15}{45}-\frac{45}{45}\right)$

$\frac{x+60}{31}-\frac{x+60}{33}=\frac{x+60}{43}-\frac{x+60}{45}$

$\frac{x+60}{31}-\frac{x+60}{33}-\frac{x+60}{43}+\frac{x+60}{45}=0$

$(x+60).\left(\frac{1}{31}-\frac{1}{33}-\frac{1}{43}+\frac{1}{45}\right)=0$

$\Rightarrow x+60=0$ vì $\frac{1}{31}-\frac{1}{33}-\frac{1}{43}+\frac{1}{45}\neq 0$

d) $\dfrac{1}{x(x+1)}+\dfrac{1}{(x+1)(x+2)}+\dfrac{1}{(x+2)(x+3)}-\dfrac{1}{x}=\dfrac{1}{2010}$ https://youtu.be/qTGVCNgPWS0

$x=-60$

$\frac{1}{x}-\frac{1}{x+1}+\frac{1}{x+1}-\frac{1}{x+2}+\frac{1}{x+2}-\frac{1}{x+3}-\frac{1}{x}=\frac{1}{2010}$

$-\frac{1}{x+3}=\frac{1}{2010}$

$\frac{-1}{x+3}=\frac{-1}{-2010}$

$\Rightarrow \quad x+3=-2010$

$x=-2013$